

Social Media Interactions Dataset

Dataset:

Before Data Cleaning	3,200 records and 6 columns.
After Data Cleaning	3,020 records and 6 columns.

Observation	Action Taken	Rationale
180 rows of duplicate records were found.	Dropping/removing the duplicates from the dataset	Having totally identical records does not provide additional info and if left in, it could lead to inaccuracy in analysis.
Platform column contained missing values.	Replaced the missing values as 'undetermined'	Determining the actual platform from the data is not possible, using the mode is not applicable, as this information won't be supported and accurate.
Sentiment column contained missing values.	Replaced the missing values as 'undetermined'	Same as Platform, determining the Sentiment is not possible through mode as it does not guarantee accuracy about the actual satisfactory level of the customer. Leaving it undetermined rather than assuming their Sentiment.
InteractionDate was stored as an object data type and was in mixed date formats.	Standardized dates from into a single datetime format. 'YYYY-MM-DD'	Consistency is valuable especially for chronological or time-based analysis.

Customer Demographics Dataset

Dataset:

Before Data Cleaning	3,200 records and 6 columns.
After Data Cleaning	3,023 records and 6 columns.

Observation	Action Taken	Rationale
177 rows of duplicate records were found.	Dropping/removing the duplicates from the dataset	Having totally identical records does not provide additional info and will cause the results in customer to be counted multiple times.
'Age' was stored as object	Converted 'Age' to numerical data type.	Age is a numerical variable and needs to be numeric for analysis.
'Age' column contained problematic values	Converted the values to missing during numeric conversion Values below 0 and above 100 were converted to missing, used the median age of 45 to fill missing/invalid ages as the valid ages in were determined to be '18-70'	The actual age could not be determined. So median was selected because it is less sensitive to extreme values and produces a reasonable representative age base on the dataset.
'IncomeLevel' contained missing values	Replaced missing values with 'Undetermined'	Income level could not reliably be inferred from the available data. "Undetermined" makes this distinction explicit instead of incorrectly assigning the most common income category.

'SignUpDate' was stored as an object data type and was in mixed date formats.	Standardized dates from into a single datetime format. 'YYYY-MM-DD'	Consistency is valuable especially for chronological or time-based analysis.
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Customer Transactions Dataset

Dataset:

Before Data Cleaning	3,200 records and 6 columns.
After Data Cleaning	3,015 records and 6 columns.

Observation	Action Taken	Rationale
185 duplicate rows were found.	Dropping/removing the duplicates from the dataset	Having totally identical records does not provide additional info and will cause the results in transactions to be counted multiple times.
'Amount' column contained several problematic values	'Free' values converted to 0 Negative values are converted as missing Missing values are then replaced with the median amount	Free value could equate as no monetary charged, so it was interpreted as a zero-value transaction. Negative values like -100 is an invalid transaction and could not be determined why. So it was treated as missing so it would not distort numerical analysis. Median was selected because it is less affected by unusually high or low transaction amounts than if the Mean was used.
'Amount' was stored as object	Converted 'Amount' to numerical data type	Converted to proceed with the calculations to get the Median.
'ProductCategory' column contained missing values	Replaced missing values with 'Undetermined'	Actual product category could not be determined, so assigning a specific category

		would introduce unsupported assumptions. <i>*ProductCategory could be determine for missing values if similar 'Amount' could be linked to a ProductCategory (recheck this one laturrr)</i>
'TransactionDate' was stored as an object data type and was in mixed date formats.	Standardized dates from into a single datetime format. 'YYYY-MM-DD'	Consistency is valuable especially for chronological or time-based analysis.